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## Report – Mini Project 2

1. For every system call, changes have been made to the following files.

- “kernel/syscall.h”
- “kernel/syscall.c”
- “kernel/usys.S”
- “user/user.h”
- “kernel/sysfile.c” / “kernel/sysproc.c”

All the helper functions have been included in “proc.c” to implement the system call “getsyscount” a separate file “user/syscount.c” has been included.

```
int get_syscall_count(int mask)
{
    struct proc *p = myproc();
    int syscall_num = 0;

    while (mask > 1)
    {
        mask >>= 1;
        syscall_num++;
    }

    return p->syscall_count[syscall_num];
}
```

Ques. What is the implication of adding the arrival time in the lottery based scheduling policy? Are there any pitfalls to watch out for? What happens if all processes have the same number of tickets?

Ans. Adding arrival time in LBS, helps in considering the one which has arrived earlier in case 2 processes have same tickets. The one waiting earlier gets implemented first. Yes there are pitfalls to watch out for. Newer processes getting priority will not allow older processes to run leading to starvation. If all the newer processes have higher priority. If all the processes have same number of tickets, arrival time is considered to choose the process.

